

GPU-Accelerated Continuous-Time Successive Convexification for Contact-Implicit Legged Locomotion

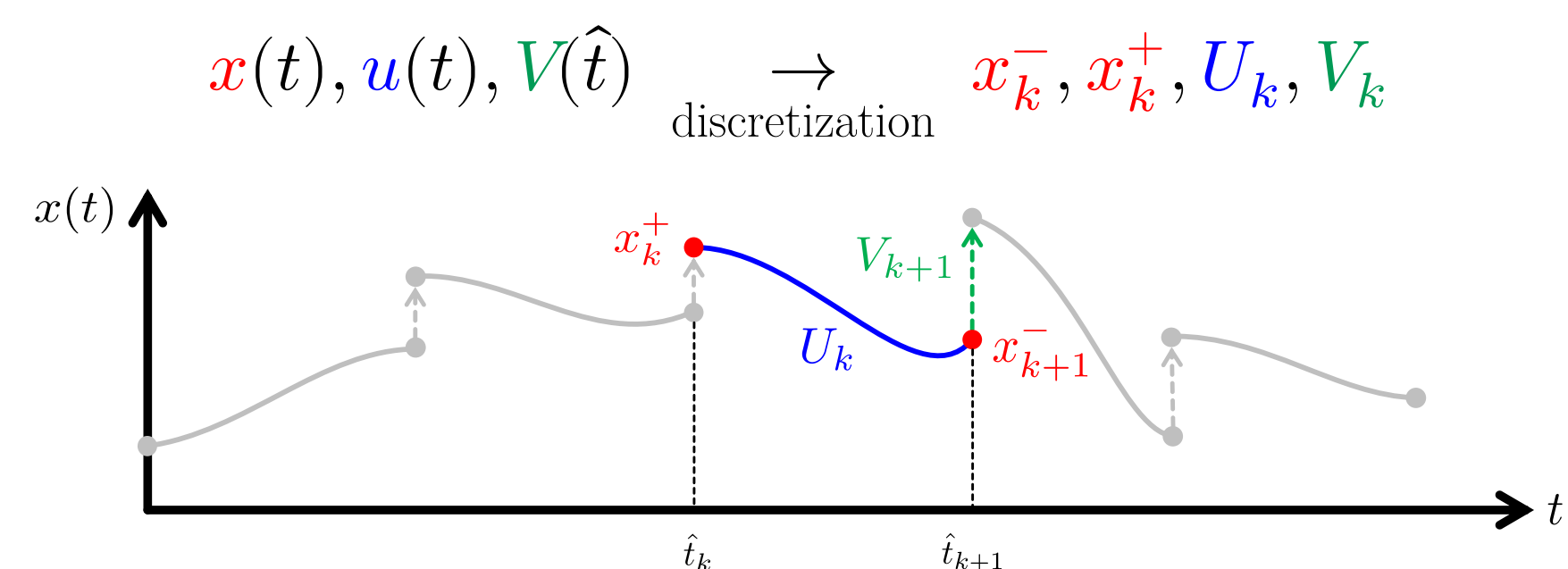
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Hybrid Optimal Control

We model optimal control problems using **direct multiple-shooting**, with impulses allowed to occur at candidate impact times $\hat{t} \in [0, t_f]$.

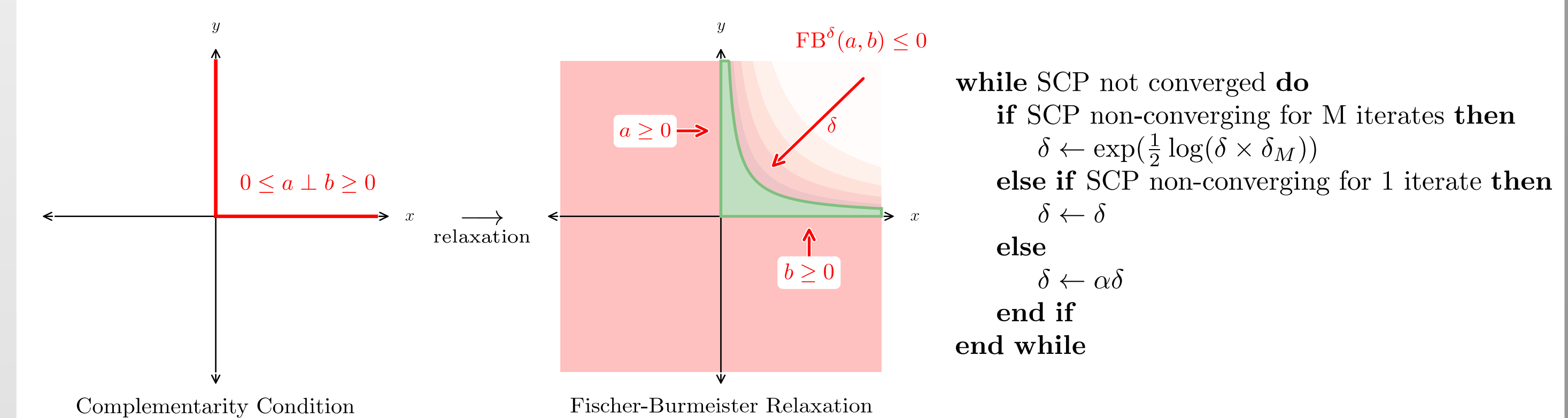


ci-SCvx: Contact-Implicit Successive Convexification

We introduce the **contact-implicit successive convexification** (ci-SCvx) framework with the following core features:

1. A sequential convex programming (SCP) algorithm with a customized GPU-accelerated implementation (Python/JAX) for fast solve times.
2. Exact continuous-time optimization over contact modalities using integral cross-complementarity constraints and time dilation.
3. A backtracking homotopy scheme within SCP iterations to progressively tighten relaxation parameters associated with complementarity conditions.

Backtracking δ -Homotopy



Contact-Implicit Dynamics

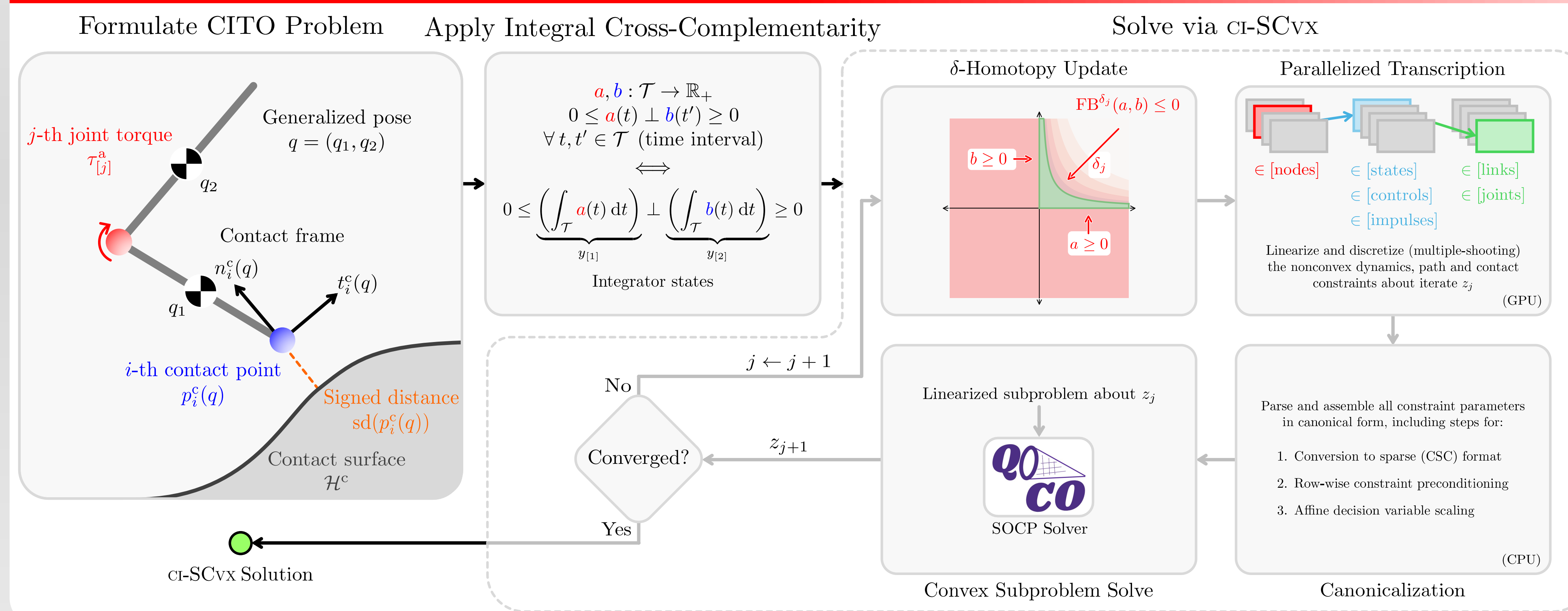
Continuous-Time Dynamics

$$\begin{aligned} \dot{q} &= v \\ \dot{v} &= M(q)^{-1} \left(W(\tau^n) + \sum_{i=1}^{n^c} J_i^c(q)^\top R_i^c(q) \Phi_i \right) \\ 0 &\leq \Phi_i^n \perp \text{sd}(p_i^c(q)) \geq 0 \\ \Phi_i^t &\in \arg\max_{\Phi} - (R_i^{c,t}(q) \Phi)^\top J_i^c(q) v \quad \text{s.t.} \quad \|\Phi\| \leq \mu_i^c \Phi_i^n \\ i &\in 1, \dots, n^c \end{aligned}$$

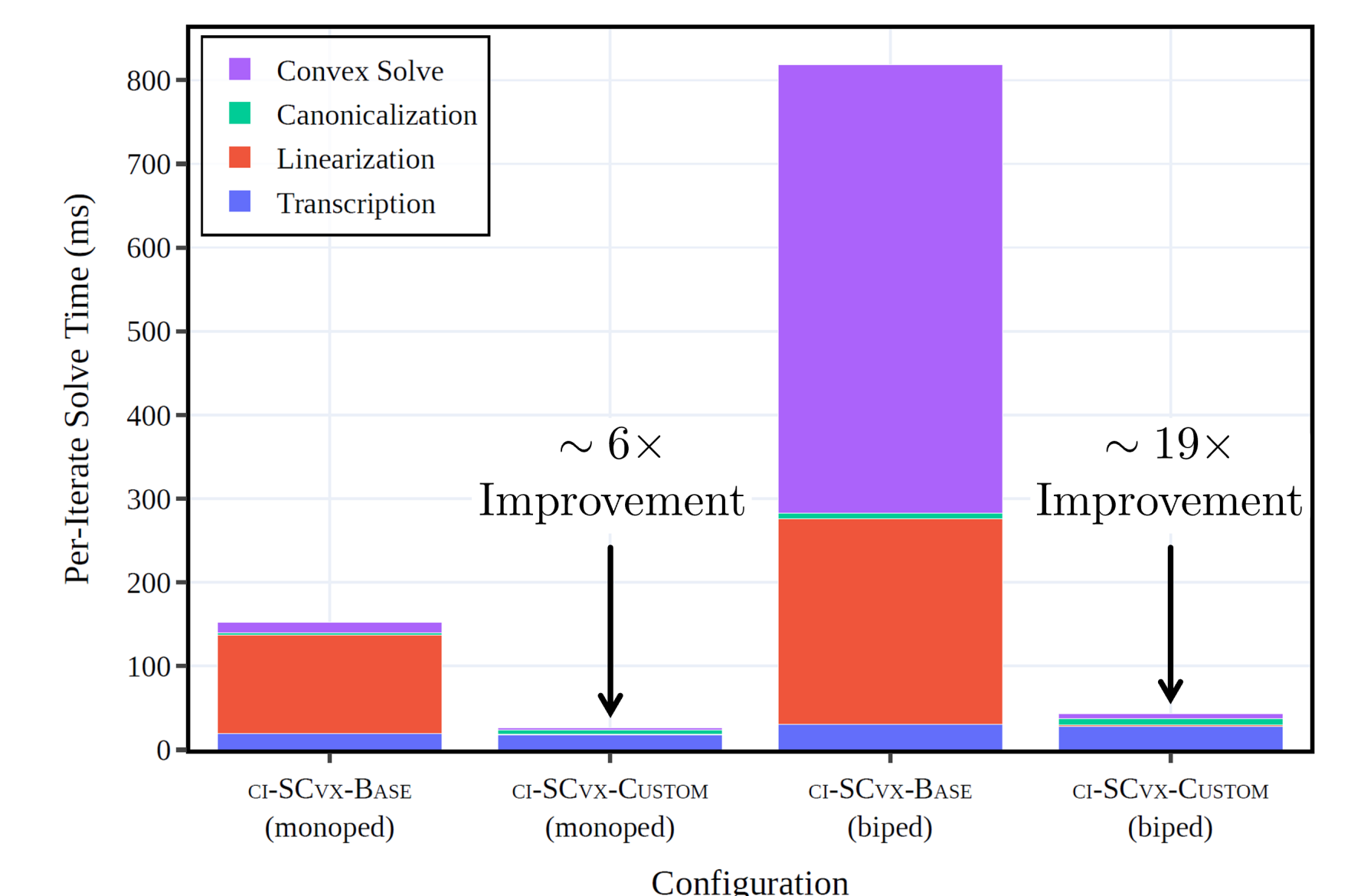
Impulse Update

$$\begin{aligned} M(q_i)(v_i^+ - v_i^-) &= \sum_{i=1}^{n^c} J_i^c(q_i)^\top R_i^c(q_i) \Phi_{i,\hat{t}}^c \\ 0 &= \Phi_{i,\hat{t}}^{c,n} R_i^{c,n}(q_i)^\top J_i^c(q_i)(v_i^+ - v_i^- - \varepsilon_i^c v_i^-) \\ 0 &\leq \Phi_{i,\hat{t}}^{c,n} \perp \text{sd}(p_i^c(q_i)) \geq 0 \\ \Phi_{i,\hat{t}}^{c,t} &\in \arg\max_{\Phi} - (R_i^{c,t}(q_i) \Phi)^\top J_i^c(q_i) v_i^+ \quad \text{s.t.} \quad \|\Phi\| \leq \mu_i^c \Phi_{i,\hat{t}}^{c,n} \\ \hat{t} &\in 1, \dots, N-1, \quad i \in 1, \dots, n^c \end{aligned}$$

Architecture of ci-SCvx



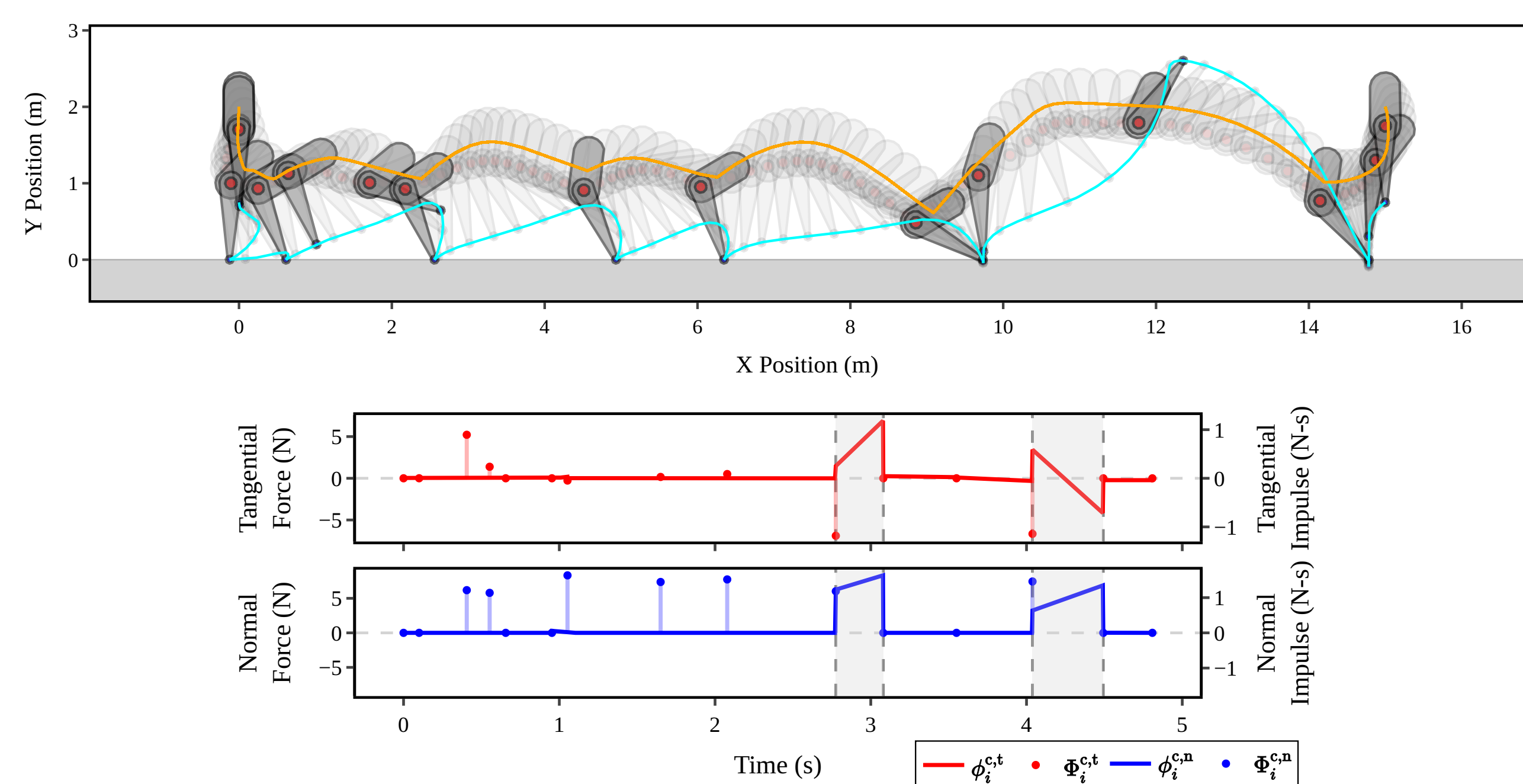
Custom Implementation Speedups



We accelerate solve times relative to existing SOTA primarily due to two improvements:

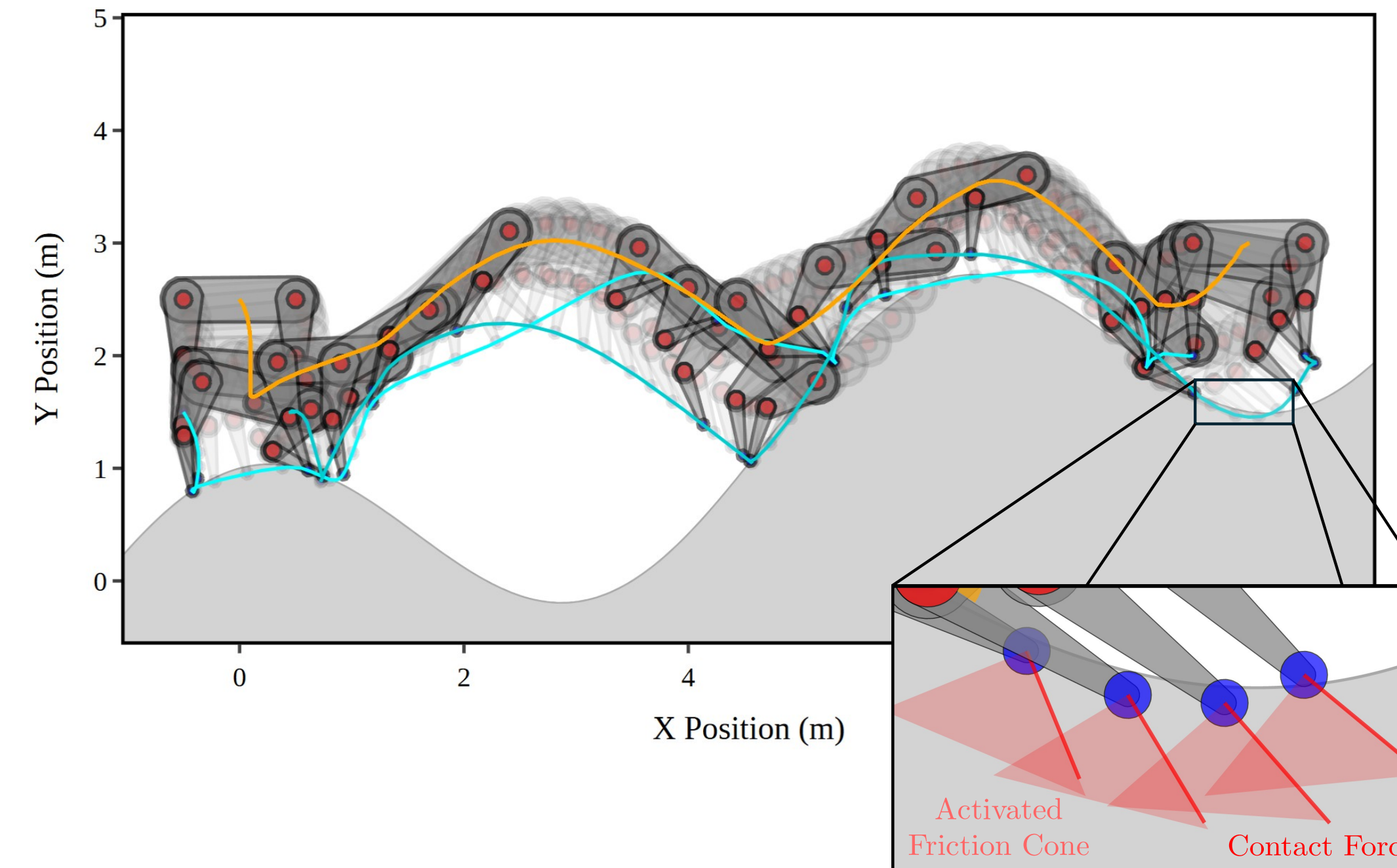
1. Sparsity-exploiting custom convex subproblem parser.
2. Batch linearization across all nonconvex constraints.

Case Study: Monoped



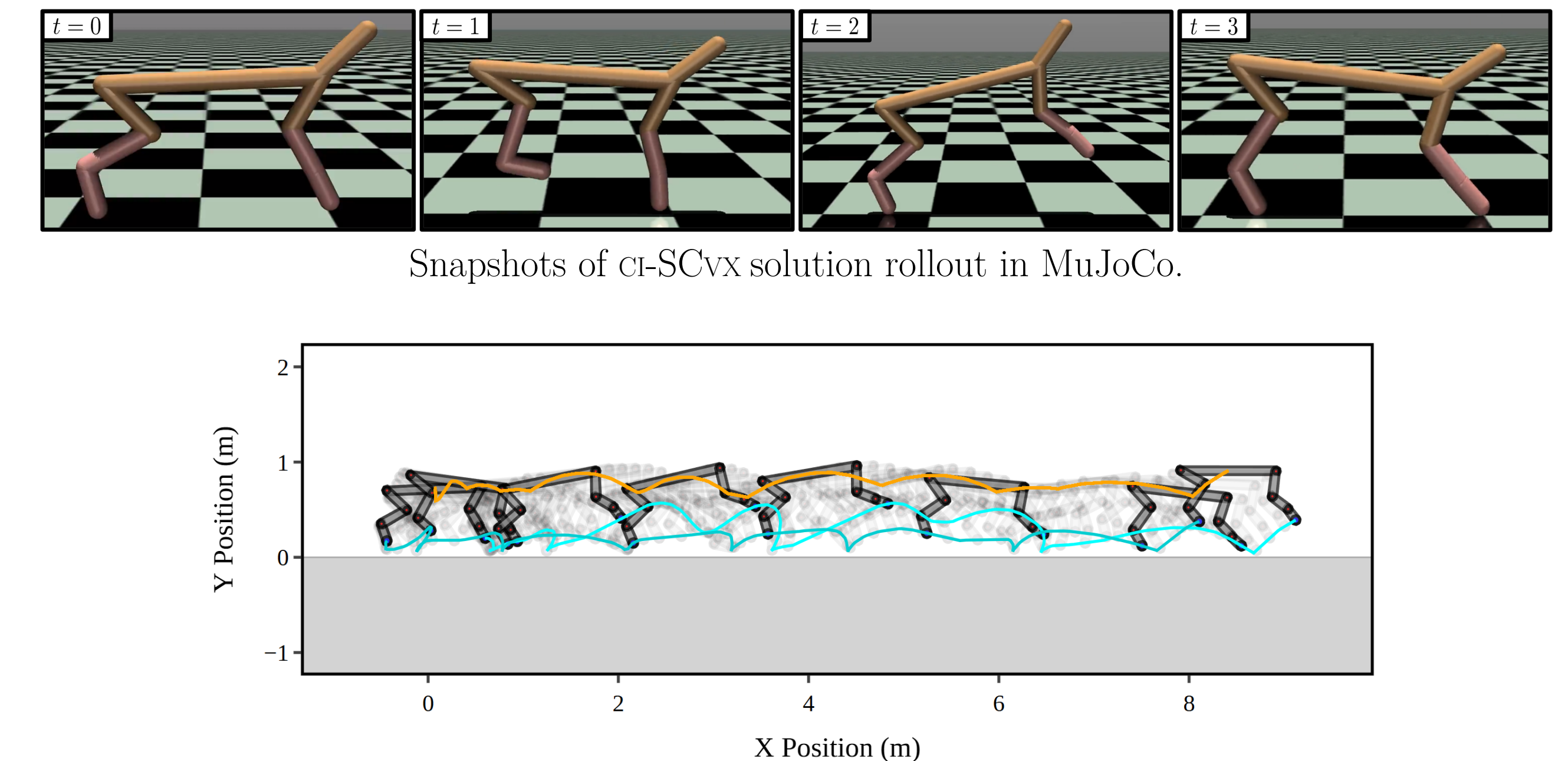
Monoped ci-SCvx solution with a combination of impulsive and persistent sticking contact.

Case Study: Biped



Biped ci-SCvx solution on a sinusoidal surface with slipping behavior observed.

Benchmarking: Half-Cheetah (MuJoCo)



Half-Cheetah ci-SCvx solution tracked with MuJoCo-MPC; we observe a $\sim 7\%$ improvement relative to baseline (standalone MuJoCo-MPC) in terms of Gymnasium environment objective.